

Inner Product Masked Integral Distinguishers and Integral Sets over Large Finite Fields

Applications to MiMC, CIMINION and Chaghri

Weizhe Wang, Deng Tang^(✉), and Haoyang Wang

Shanghai Jiao Tong University, Shanghai 200240, China
{SJTUwzz, dengtang, haoyang.wang}@sjtu.edu.cn

Abstract. The security and performance of many advanced protocols heavily rely on the underlying symmetric-key primitives. These primitives, known as arithmetization-oriented (AO) ciphers, focus on arithmetic metrics over large finite fields. Most AO ciphers are vulnerable to algebraic attacks, especially integral attacks. In this work, we explore integral attacks over large finite fields. By combining integral distinguishers with inner product masks, we propose inner product masked (IPM) integral distinguishers and establish a system of equations concerning the inner product mask. Additionally, we provide theoretical lower bounds on the complexity of IPM integral distinguishers in certain cases. For practical applications, we introduce IPM integral sets, which effectively characterize the integral property of sets over the finite field $\mathbb{F}_{p^n}$. Besides the IPM sets based on additive subgroups and multiplicative subgroups, we present a method to obtain sets with improved integral properties by merging existing sets. Exploring different classes of IPM integral sets can help us find lower-complexity integral distinguishers. Combining with monomial detection techniques, we propose a framework for searching for integral distinguishers based on the IPM integral set. Our framework is compatible with various monomial detection techniques, including general monomial prediction proposed by Cui et al. at ASIACRYPT 2022 and coefficient grouping invented by Liu et al. at EUROCRYPT 2023. Previous integral distinguishers over $\mathbb{F}_{2^n}$ were predominantly based on additive subgroups. With IPM integral sets based on multiplicative subgroups and merged sets, we successfully obtain IPM integral distinguishers with lower complexity for MiMC, CIMINION, and Chaghri. We believe that our work will provide new insights into integral attacks over large finite fields.

Keywords: Integral attacks · IPM integral distinguishers · IPM integral sets · Multiplicative subgroups · Monomial detection

1 Introduction

In recent years, symmetric-key cryptographic primitives, such as stream ciphers, block ciphers, hash functions, pseudorandom generators, etc., serve as fundamen-

tal building blocks for many advanced protocols like secure multi-party computation (MPC), fully homomorphic encryption (FHE) and zero-knowledge proof systems (ZK). Traditionally, linear and nonlinear operations in symmetric-key primitives have similar computational costs. However, in MPC/FHE/ZK protocols, nonlinear operations like multiplications are more expensive and often become bottlenecks. Consequently, traditional primitives like AES are inefficient in these contexts, necessitating the design of symmetric-key primitives specifically optimized for advanced cryptographic protocols. Since the invention of block cipher LowMC [2], a large number of MPC/FHE/ZK-friendly symmetric-key primitives have been developed. Among them, some innovative primitives that prioritize arithmetic metrics are known as arithmetization-oriented (AO) ciphers. In contrast to using small S-boxes as the traditional ones, AO ciphers are preferred to employ nonlinear functions with explicit and concise algebraic representations over large finite fields, e.g., MiMC [1], CIMINION [10], and Chaghri [3] are three classical AO ciphers. MiMC is a block cipher with extremely low multiplicative complexity proposed at ASIACRYPT 2016. Since its birth, MiMC has gained widespread attention. Due to the algebraic simplicity of the round function, MiMC has become the preferred choice for many use cases [13, 23]. It also served as a candidate for Zcash¹ and a baseline in the “STARK-Friendly Hash Challenge” competition. Recently, MiMC has been applied to zkBridge [29], which is an efficient cross-chain bridge proposed at ACM CCS 2022. CIMINION is another MPC-friendly encryption scheme proposed at EUROCRYPT 2021. Thanks to Toffoli gates and Farfalle-like construction, CIMINION requires an extremely low average number of field multiplications when encrypting a large number of plaintexts. Chaghri, proposed at ACM CCS 2022, is an FHE-friendly block cipher defined over the finite field $\mathbb{F}_{2^{63}}$ with efficient circuit implementation. Its implementation with HELib can outperform AES by about 65%.

The structure of AO ciphers has a great influence on the type of attacks that can be mounted. Statistical attacks do not appear to pose a significant threat to the security of AO ciphers due to their wide trail strategy. However, algebraic attacks, especially higher-order differential attacks [5, 6, 8, 11, 20, 21] are highly effective against on AO ciphers. Higher-order differential attacks are essentially a specific variant of integral attacks based on additive subgroups (or vector subspaces). There have been numerous research achievements and well-established research methodologies for integral attacks, most of which focus on block ciphers over $\mathbb{F}_2^n$. For block ciphers over finite fields with characteristics other than two, Beyne et al. [5] introduced the use of multiplicative subgroups and successfully obtained zero-sum distinguishers on block ciphers when the degree of the cipher is less than the size of subgroups. While there is some ongoing research and application of integral attacks over large finite fields, there is still a lack of universal tools and frameworks like division property [14, 26, 27], which greatly hinders the development of integral attacks. Furthermore, in previous research on integral attacks, some researchers have attempted to conduct public operations on plaintexts or ciphertexts before summing them up [15, 16, 19, 28].

¹ <https://github.com/zcash/zcash/issues/2233>

These findings suggest that these operations contribute to strengthening the impact of integral attacks. To the best of our knowledge, this enhancement also has only been applied to integral attacks over $\mathbb{F}_2^n$. In this regard, we would like to underline that further research in this domain for ciphers over $\mathbb{F}_{2^n}$ and $\mathbb{F}_p$ needs additional attention.

1.1 Our Contribution and Organization

In this paper, we conduct in-depth research on integral attacks over large finite fields. In Section 2, we introduce the notations and some background knowledge about finite fields, integral attacks, and the monomial detection. As mentioned above, the existing integral attacks are simply based on the degree or algebraic degree of the cipher to form distinguishers, which is a rather crude approach. It is more appropriate to explore various structures of input sets according to the polynomial of target cipher, as this can help us construct distinguishers with lower complexity, thereby enhancing the effectiveness of integral attacks. The contributions of this paper are summarized below.

1. A generalized version of integral distinguishers, called inner product masked (IPM) integral distinguishers, is introduced in Section 3. The task of constructing IPM integral distinguishers is reformulated as solving a system of linear equations under certain reasonable assumptions in Remark 1. Following that, we present theoretical lower bounds on the complexity of IPM integral distinguishers. In particular, we derive a lower bound of $3^r + 2$ for r -round MiMC- p/p according to Proposition 1. In Remark 2, we also validate the importance of strong linear layers in block ciphers.
2. In Section 4, we propose a new concept called IPM integral sets to better characterize the integral property of a given set over large finite fields. It can be seen as a generalization of three-subset division property without unknown set [14]. Additive subgroups and multiplicative subgroups are two different classes of IPM integral sets. Moreover, we present an approach for merging existing sets to create a new set with improved integral property. Furthermore, a framework for determining integral distinguishers using monomial detection techniques is proposed as Algorithm 1.
3. We utilize two advanced monomial detection techniques, specifically general monomial prediction and coefficient grouping, and apply our framework to MiMC, CIMINION and Chaghri in Section 5. With IPM integral sets based on multiplicative subgroups and merged sets, we obtain some distinguishers with lower complexity compared to previous ones [8, 20, 30]. A selection of our result is displayed in Table 1.

Based on the above results, we believe that our framework is useful for carrying out integral attacks over large finite fields and is worth further investigation. One may ask in this regard, that the improvements do not look very significant in terms of numerical values. Here, it should be clarified that this work mainly fills the gap in integral attacks over large finite fields, as previous research on integral

Table 1: Some integral distinguishers for different **A0** ciphers

Cipher	Round	Previous result	Source	Our result
MiMC₃ (129, r)	3	2^5	ASIACRYPT 2022 [8]	2^3
	7	2^{12}		$2^{10.33}$
	60	2^{95}		$2^{94.75}$
MiMC₅ (129, r)	2	2^4		2^3
	20	2^{45}		$2^{44.59}$
MiMC₇ (129, r)	17	2^{46}		$2^{45.81}$
MiMC₉ (129, r)	2	2^4		2^3
MiMC₁₅ (129, r)	31	2^{121}		$2^{120.25}$
MiMC₁₇ (129, r)	2	2^4		2^3
	12	2^{45}		2^{43}
FeistelMiMC₃ (129, r)	7	2^{12}		$2^{11.56}$
	74	2^{118}		$2^{117.74}$
CIMINION	23	2^{23}	INSCRYPT 2022 [30]	$2^{21.42}$
	39	2^{39}		$2^{37.42}$
	63	2^{63}		$2^{61.5}$
Chaghri	4	2^{18}	EUROCRYPT 2023 [20]	$2^{17.39}$
	8.5	2^{41}		$2^{40.88}$

attacks was mostly focused on $\mathbb{F}_2$. According to this initiative, an attacker can choose a more appropriate input set to obtain a distinguisher with lower complexity. Two very recent works [8, 20] provide the monomial detection methods and our framework is compatible with any such monomial detection techniques. We can achieve better results than [8, 20], which are the SOTA contributions in recent times, because we choose multiplicative subgroups and merged sets as inputs, while the works in [8, 20] only exploit additive subgroups (or vector subspaces) as inputs. As we like to take the question of modest complexity improvements in terms of numerical results at the same number of rounds (mostly less than half, sometimes up to half, or even a quarter) compared to [8, 20], the small improvements are attributed to the highly precise monomial detection technique in [8, 20]. Compared to other works (e.g., results on CIMINION in [30]), we can achieve at least a halving of complexity at most number of rounds (as shown in Table 7). In this context, we argue that such improvements could not be achieved without the novel techniques mentioned in this paper, and all of our identified distinguishers are new. The existing results are already very competitive and we note that the straightforward progress of the previous techniques cannot obtain the improved results we achieve here.

We conclude the paper in Section 6. Several explanations are added in the appendices, which may be referred to if additional clarifications are required. The code of our experiments is available at Github repository. Our experiments are implemented on the Intel Xeon Gold 5220R CPU 2.2 GHz.

2 Preliminaries

In this paper, we will use the following notations. We denote a finite field with q elements as $\mathbb{F}_q$. We denote $\mathbb{F}_q^* = \mathbb{F}_q \setminus \{0\}$. Let $\mathbb{F}_{p^n}$ denote the finite field with characteristic p and $\mathbb{F}_{p^n}^t$ denote the t -dimensional vector space over the finite field $\mathbb{F}_{p^n}$. We denote the polynomial ring over $\mathbb{F}_{p^n}$ as $\mathbb{F}_{p^n}[x]$ and the set of integers $\{i : a \leq i \leq b\}$ as $[a, b]$. For a set $\mathbb{X}$, we use $|\mathbb{X}|$ to represent the number of elements. The set $\{i : i \in A, i \notin B\}$ is denoted by $A - B$. We use bold italic letters to represent vectors, e.g. $\mathbf{a} \in \mathbb{F}_q^t$ denotes the vector $\mathbf{a} = (a_0, a_1, \dots, a_{t-1})$, where $a_i \in \mathbb{F}_q$. For any $a \in [0, 2^n - 1]$, we have $a = \sum_{i=0}^{n-1} a_i \cdot 2^i$, $a_i \in \{0, 1\}$, and its Hamming weight is $wt(a) = |\{i : a_i \neq 0\}|$. Correspondingly, the Hamming weight of $\mathbf{u} \in [0, 2^n - 1]^t$ is $wt(\mathbf{u}) = \sum_{i=0}^{t-1} wt(u_i)$. We use $\oplus$ as addition over $\mathbb{F}_2$ or $\mathbb{F}_{2^n}$ and $+$ over $\mathbb{F}_p$ or $\mathbb{F}_{p^n}$. For simplicity, we denote a set of all integers $j \in [0, n-1]$ satisfying $d \mid j$ as D_d^n and a set of all integers $j \in [0, n-1]$ satisfying $d \nmid j$ as UD_d^n , i.e., $D_d^n = \{j \in [0, n-1] : d \mid j\}$, $UD_d^n = \{j \in [0, n-1] : d \nmid j\}$. We use $a \% b$ to represent $a \bmod b$.

Let $F : \mathbb{F}_{p^n}^t \rightarrow \mathbb{F}_{p^n}$ be a function over $\mathbb{F}_{p^n}[x_0, x_1, \dots, x_{t-1}]/(x_0^{p^n} - x_0, x_1^{p^n} - x_1, \dots, x_{n-1}^{p^n} - x_{n-1})$. Function F can be uniquely expressed as

$$F(\mathbf{x}) = F(x_0, \dots, x_{t-1}) = \sum_{\mathbf{u} \in \{0, \dots, p^n-1\}^t} a_{\mathbf{u}} \mathbf{x}^{\mathbf{u}} = \sum_{i_0=0}^{p^n-1} \cdots \sum_{i_t=0}^{p^n-1} a_{i_0, \dots, i_t} x_0^{i_0} \cdots x_t^{i_t},$$

where every coefficient $a_{\mathbf{u}} \in \mathbb{F}_{p^n}$, and $\mathbf{x}^{\mathbf{u}} = \pi_{\mathbf{u}}(\mathbf{x}) = \prod_{i=0}^{n-1} x_i^{u_i}$ is called a monomial. If the coefficient of $\mathbf{x}^{\mathbf{u}}$ is non-zero, it means that $\mathbf{x}^{\mathbf{u}}$ is contained by F , denoted by $\mathbf{x}^{\mathbf{u}} \rightarrow F$. The univariate degree of F on variable x_j is denoted by $deg_j(F)$ and defined as:

$$deg_j(F) = \max\{i_j : 0 \leq i_j \leq p^n - 1, a_{i_0, i_1, \dots, i_t} \neq 0\}.$$

Moreover, when $p = 2$, the algebraic degree $\delta(F)$ can be computed as

$$\delta(F) = \max\left\{\sum_{j=1}^t wt(i_j) : 0 \leq i_j \leq 2^n - 1, a_{i_0, i_1, \dots, i_t} \neq 0\right\}.$$

2.1 Finite Fields

The finite field consists of a finite set of elements along with two operations, addition and multiplication. $\mathbb{F}_p$ with a large prime number p and $\mathbb{F}_{2^n}$ are widely used in cryptography. The multiplicative group of a finite field is the group under multiplication concerning this finite field. A useful lemma for multiplicative subgroups is given below [22].

Lemma 1. *The multiplicative group $\mathbb{F}_q^*$ of a finite field $\mathbb{F}_q$ is cyclic. Let H be a cyclic group generated by x , if $|H| = n$ is finite, then for each positive integer $a \mid n$, there is a unique subgroup of H of order a . This subgroup is the cyclic group $\langle x^d \rangle$, where $d = \frac{n}{a}$.*

2.2 Integral Attacks

In an integral attack, the most crucial aspect is to find an input set for which the sum of their outputs is a key-independent constant value. Such an input set is called an *integral distinguisher*. A concrete definition of an integral distinguisher is given below.

Definition 1 (Integral Distinguisher). *Let $E_k^r : \mathbb{F}_{p^n} \rightarrow \mathbb{F}_{p^n}$ be an r -round block cipher and $\mathbb{X} \subseteq \mathbb{F}_{p^n}$. If the sum of ciphertexts over $\mathbb{X}$ satisfies*

$$\sum_{x \in \mathbb{X}} E_k^r(x) = c,$$

where c is a constant over $\mathbb{F}_{p^n}$ that does not depend on the secret key k , then $\mathbb{X}$ is an integral distinguisher of the block cipher. Generally, we have $c = 0$.

In previous bit-based integral attacks, the input set usually consists of plaintexts taking all possible combinations in d input positions, whereas the remaining bits take a fixed value. According to the theory of higher-order differential cryptanalysis [18], the sum of any Boolean function with its algebraic degree smaller than d over such a set would become zero and naturally form an integral distinguisher.

With an integral distinguisher, the attacker can recover partial information of the secret key by guessing, and then the remaining part of the key can be recovered by a brute-force attack. However, the trivial key-guessing approach is inefficient on A0 ciphers like MiMC. In [11], Eichlseder et al. proposed a new strategy for key recovery by constructing a low-degree polynomial.

The division property [14, 26, 27] is an accurate and generic tool to search for integral distinguishers over $\mathbb{F}_2^n$. In the development of the division property, various versions of definitions have emerged, each differing in terms of accuracy and efficiency. Some common versions of the division property are given in Appendix B.

2.3 Monomial Detection

Monomial detection is an important method used to identify the presence of monomials in a given function, which is crucial for estimating the (algebraic) degree of the function. Monomial prediction [17], based on the division property, is a monomial detection technique that is widely used in cube attacks and integral attacks. Later, the generalized monomial prediction [8] was proposed, and used to attack A0 ciphers like MiMC. The coefficient grouping [20, 21] is another monomial detection technique. The main idea of coefficient grouping is to

give a compact expression for all possible monomials and encode it as some constraints. Compared to monomial prediction, coefficient grouping is more efficient and lightweight. However, its applicability is somewhat limited and is currently confined to field-based block ciphers with quadratic round function.

3 IPM Integral Distinguishers

3.1 A More Generalized Integral Distinguisher

In previous integral attacks on bit-based block ciphers, attackers can perform some public mapping on the plaintexts [19, 28] and their corresponding ciphertexts [15, 16] to obtain more distinguishers. Generally, the operation performed on each element is the same. However, this is not strictly necessary; attackers could even perform various operations on different elements and then sum them up, such as multiplying ciphertexts by different constants. In particular, this multiply-and-add operation introduces an inner product mask during summation. We would like to refer to this type of distinguisher as *inner product masked (IPM) integral distinguisher*. The definition of IPM integral distinguisher is as follows.

Definition 2 (IPM Integral Distinguisher). Let $E_k^r : \mathbb{F}_{p^n} \rightarrow \mathbb{F}_{p^n}$ be an r -round block cipher and $\mathbb{X} \subseteq \mathbb{F}_{p^n}$. If there exists a vector $\mathbf{c} \in \mathbb{F}_{p^n}^{|\mathbb{X}|}$ such that

$$\mathbf{c}^T E_k^r(\mathbb{X}) = \sum_{j=1}^{|\mathbb{X}|} c_j E_k^r(x_j) = 0$$

and the sum does not depend on the secret key k , then $\mathbb{X}$ is an IPM integral distinguisher of the block cipher. $E_k^r(\mathbb{X}) \in \mathbb{F}_{p^n}^{|\mathbb{X}|}$ denotes the vector of ciphertexts.

The original integral distinguisher can be seen as a special IPM integral distinguisher whose mask is an all-one vector. With an inner product mask, it is possible to construct an integral distinguisher with lower complexity. A simple example is given in Appendix G.1. It should be noted that bit-based IPM integral distinguishers are trivial because of the limited choice of coefficients over $\mathbb{F}_2$.

3.2 Properties of IPM Integral Distinguishers

Consider an r -round block cipher $E_k^r : \mathbb{F}_{p^n} \rightarrow \mathbb{F}_{p^n}$ with a secret key $k \in \mathbb{F}_{p^m}$, we can represent its function as

$$E_k^r(x) = \sum_{i=0}^{p^n-1} a_i(k) x^i,$$

where $a_i(k)$ is a function $a_i : \mathbb{F}_{p^m} \rightarrow \mathbb{F}_{p^n}$ mapping a key to a value in $\mathbb{F}_{p^n}$. Let I be the set of indices satisfying $a_i(k) \neq 0$ for $i \in [0, p^n - 1]$. Given a set of plaintexts $\mathbb{X}$, the sum of ciphertexts is

$$\sum_{j=1}^{|\mathbb{X}|} c_j E_k^r(x_j) = \sum_{j=1}^{|\mathbb{X}|} \sum_{i=0}^{p^n-1} c_j a_i(k) x_j^i = \sum_{j=1}^{|\mathbb{X}|} \sum_{i \in I} c_j a_i(k) x_j^i = \sum_{i \in I} a_i(k) \sum_{j=1}^{|\mathbb{X}|} c_j x_j^i.$$

A sufficient condition for the sum to be zero is $\sum_{j=1}^{|\mathbb{X}|} c_j x_j^i = 0$ for all $i \in I$.

Remark 1. It is possible for different terms $a_i(k)$ to cancel each other out, which means they are linearly dependent. However, this case is hard to achieve. Therefore, we would like to assume that different terms $a_i(k)$ are all linearly independent. Moreover, we also assume that $a_0(k)$ always be non-zero.

Suppose $|\mathbb{X}| = t$, then a system of equations for IPM integral distinguishers can be obtained under the assumption.

$$\begin{cases} c_1 + c_2 + \dots + c_t = 0 \\ c_1 x_1^{i_1} + c_2 x_2^{i_1} + \dots + c_t x_t^{i_1} = 0 \\ \dots \\ c_1 x_1^{i_{|I|-1}} + c_2 x_2^{i_{|I|-1}} + \dots + c_t x_t^{i_{|I|-1}} = 0 \end{cases} \quad (1)$$

Equation (1) can also be written as an equation $M\mathbf{c} = \mathbf{0}$, where

$$M = \begin{pmatrix} 1 & 1 & \dots & 1 \\ x_1^{i_1} & x_2^{i_1} & \dots & x_t^{i_1} \\ \vdots & \vdots & & \vdots \\ x_1^{i_{|I|-1}} & x_2^{i_{|I|-1}} & \dots & x_t^{i_{|I|-1}} \end{pmatrix} \text{ and } \mathbf{c} = (c_1, c_2, \dots, c_t)^T.$$

Now, the problem is equivalent to solving a linear equation. The coefficient matrix M is determined jointly by $\mathbb{X}$ and the index set I . In the case of a specific block cipher E_k^r , the non-zero coefficient index set I is fixed. Attackers need to select a proper plaintext set $\mathbb{X}$ such that the equation $M\mathbf{c} = \mathbf{0}$ has a non-zero solution. To lower the complexity of attacks, attackers typically prefer using an $\mathbb{X}$ as small as possible.

Denote the rank of matrix M as $R(M)$. The number of solutions to the equation $M\mathbf{c} = \mathbf{0}$ is closely related to $R(M)$. For a given $|I|$ -by- t matrix M , its rank $R(M)$ is always less than or equal to $|I|$ and t , i.e. $R(M) \leq \min\{|I|, t\}$. Therefore, we have $R(M) < t$ in cases where $t > |I|$, indicating the existence of non-zero solutions to the equation. Based on this fact, we can easily derive a useful theorem.

Theorem 1. *Let $E_k^r : \mathbb{F}_{p^n} \rightarrow \mathbb{F}_{p^n}$ be an r -round block cipher with $E_k^r(x) = \sum_{i=0}^{p^n-1} a_i(k) x^i$ and $I = \{i : a_i(k) \neq 0\}$, there must exist an IPM integral distinguisher with $|I| + 1$ elements.*

Theorem 1 is a general theorem for IPM integral distinguishers. When the number of non-zero terms $|I|$ is small, attackers are able to construct an IPM integral distinguisher with low complexity. It also indicates that attackers can always find an IPM integral distinguisher whose complexity is close to the number of non-zero terms in the block cipher's function.

A question that naturally arises is whether there exists an IPM integral distinguisher with lower complexity than the one provided in Theorem 1. The answer is yes in most cases. For a given I , if attackers can carefully choose a set of plaintexts $\mathbb{X}$ such that $R(M) < |\mathbb{X}| \leq |I|$, then a more compact distinguisher can be obtained. An example of smaller IPM integral distinguishers is given in Appendix G.2. However, how to address this issue still requires further research.

Because the existence of IPM integral distinguishers is highly associated with the rank of matrix $R(M)$, a few useful propositions can be obtained in some special cases. Suppose that all monomials appear in the block cipher, the matrix M would become a Vandermonde matrix. In this case, a lower bound about the complexity of IPM integral distinguishers can be derived.

Proposition 1. *Let $E_k^r : \mathbb{F}_{p^n} \rightarrow \mathbb{F}_{p^n}$ be an r -round block cipher with $E_k^r(x) = \sum_{i=0}^{p^n-1} a_i(k)x^i$ and $I = \{i : a_i(k) \neq 0\}$, if $\deg(E_k^r(x)) = d$ and $I = \{0, 1, 2, \dots, d\}$, then constructing an IPM integral distinguisher requires at least $d + 2$ elements.*

The proof of Proposition 1 is given in Appendix A.1. Proposition 1 demonstrates that when the degree of a block cipher is high and all terms are present in its polynomial, the block cipher will become resistant to integral attacks. For the general case, we are also able to derive some valuable insights by studying the rank of the matrix.

When sorting the elements of an index/exponent set I in ascending order, the sorted I is composed of multiple contiguous sequences. The maximum length of these contiguous sequences can also serve as a measure of the block cipher's resistance against IPM integral attacks.

Proposition 2. *Let $E_k^r : \mathbb{F}_{p^n} \rightarrow \mathbb{F}_{p^n}$ be an r -round block cipher with $E_k^r(x) = \sum_{i=0}^{p^n-1} a_i(k)x^i$ and $I = \{i : a_i(k) \neq 0\} = \{i_1, i_2, \dots, i_{|I|}\}, i_1 < i_2 < \dots < i_{|I|}$, if the maximum length of I 's contiguous sequences is m , then constructing an IPM integral distinguisher requires at least $m + 1$ elements.*

The proof of Proposition 2 is given in Appendix A.2.

Remark 2. Based on the aforementioned propositions, to resist integral attacks, a block cipher must not only possess a high degree but also ensure that the number of its monomials is as large as possible. Despite restrictions on the number of columns, a large count of non-zero terms $|I|$ generally results in a high rank for the coefficient matrix M [9]. In other words, the greater the number of terms, the more difficult it becomes to achieve $t > R(M)$, making it more challenging to construct an IPM integral distinguisher and enhancing the ability of block ciphers to resist integral attacks. Therefore, a strong linear layer that can increase the number of non-zero terms quickly will help the block cipher resist IPM integral attacks.

4 IPM Integral Sets

In the previous section, we explored constructing IPM integral distinguishers by solving linear equations, but this approach appears inefficient in practical attacks. Two main factors explain this inefficiency: first, existing research lacks effective methods for selecting plaintext sets that yield low-rank matrices; second, the cipher polynomial is often too complex and dense, making calculations and storage impractical. In practice, using structured plaintext sets would be more effective. In this section, we will introduce the concept of IPM integral sets and the framework for constructing integral distinguishers.

4.1 Introduction of IPM Integral Sets

As analyzed in Section 3, the sum of ciphertexts can be decomposed into the sum of several independent terms. Because the basic component of a polynomial is the monomial, and the sum of a monomial over a set is deterministic, approaching the problem from the perspective of monomials renders the entire issue clearer and more comprehensive. To conduct a more in-depth investigation, we proposed a new concept called the *inner product masked (IPM) integral set*.

Definition 3 (IPM Integral Set). *Given a set $\mathbb{X} = \{x_1, x_2, \dots, x_t\} \subseteq \mathbb{F}_{p^n}$, a set $E \subseteq [0, p^n - 1]$ and a vector $\mathbf{c} \in \mathbb{F}_{p^n}^{|\mathbb{X}|}$, if the set $\mathbb{X}$ satisfies $\sum_{j=1}^{|\mathbb{X}|} c_j x_j^i = 0$ for all $i \in E$, then we define that $\mathbb{X}$ belongs to an $(\mathbf{c}, E)$ -IPM integral set, or $\mathbb{X}$ is an $(\mathbf{c}, E)$ -IPM integral set, where E is the set indicating integral property.*

In the absence of ambiguity, $(\mathbf{1}, E)$ -IPM integral set would be abbreviated as E -IPM integral set. A direct approach to computing an E is to iterate over all monomials. An example of IPM integral sets is given in Appendix G.3.

Specifically, some sets exhibit a constant integer value in the non-zero part, and this property can be advantageous when constructing an integral distinguisher. To effectively represent such sets, we provide a concept called *IPM integral set with two levels*. The concrete definition of IPM integral set with two levels is given in Appendix B.

The IPM integral set is a general tool for the characterization of the integral property of sets. The primary distinction between IPM integral sets and the division property lies in the introduction of the inner product mask. Even in the absence of the inner product mask, the integral set can effectively encompass the concept of the division property. A more detailed comparison between IPM integral sets and division property is given in Appendix B.

The Relation between IPM Integral Sets and IPM Integral Distinguishers Consider a field-based block cipher $E_k^r : \mathbb{F}_{p^n} \rightarrow \mathbb{F}_{p^n}$ with a secret key $k \in \mathbb{F}_{p^m}$, it can be represented by $E_k^r(x) = \sum_{i \in I} a_i(k)x^i$, where $I = \{i : a_i(k) \neq 0\} = \{i : x^i \rightarrow E_k^r(x)\}$ represents the set of indices corresponding to non-zero coefficients. Given that an input set $\mathbb{X}$ is an $(\mathbf{c}, E)$ -integral set, the IPM sum of $F(x, k)$ over $\mathbb{X}$ can be divided into two part: $\sum_{i \in I \cap E} a_i(k) \sum_{j=1}^{|\mathbb{X}|} c_j x_j^i$

and $\sum_{j \in I-E} a_i(k) \sum_{j=1}^{|\mathbb{X}|} c_j x_j^i$. Because the first part is zero, the IPM sum would also become zero if $I-E = \emptyset$, i.e. $I \subseteq E$. Therefore, we can derive the following proposition.

Proposition 3. *Let $E_k^r : \mathbb{F}_{p^n} \rightarrow \mathbb{F}_{p^n}$ be the function of a field-based block cipher and $\mathbb{X} \subseteq \mathbb{F}_{p^n}$. If there exists $\mathbf{c} \in \mathbb{F}_{p^n}^{|\mathbb{X}|}$ such that $\mathbb{X}$ is an $(\mathbf{c}, E)$ -IPM integral set, and for each monomial $x^i \rightarrow E_k^r(x)$, its exponent satisfies $i \in E$, then $\mathbb{X}$ is an IPM integral distinguisher of $E_k^r(x)$.*

4.2 Two Useful Classes of IPM Integral Sets

In this section, we will introduce two useful classes of IPM integral sets: one based on additive subgroups and the other based on multiplicative subgroups. The different structures of these two classes of IPM integral sets result in completely different integral properties.

IPM integral sets based on additive subgroups or vector subspaces are the most classical structured set for integral attacks. Its integral property is highly related to the Hamming weight of exponents. Therefore, it is powerful for some block ciphers with a low algebraic degree. In particular, for a vector subspace over $\mathbb{F}_{2^n}$, if its size is 2^d , then it would be a $\{i \in [0, 2^n - 1] : wt(i) < d\}$ -IPM integral set. Extensive research has been conducted on utilizing additive subgroups over $\mathbb{F}_2$ as integral distinguishers. Previous integral attacks on ciphers like MiMC, CIMINION and Chaghri have employed this class of integral set [8, 20, 30].

The second class of IPM integral sets is based on multiplicative subgroups. Compared to additive subgroups, multiplicative subgroups have only recently been proposed for constructing integral distinguishers, primarily over finite fields with odd characteristics. In [5], authors give the following proposition and employ multiplicative subgroups to construct zero-sum distinguishers on GMiMC.

Proposition 4. *Let G be a multiplicative subgroup of $\mathbb{F}_p^*$, where p is a prime. For any function $F : \mathbb{F}_p \rightarrow \mathbb{F}_p$ such that $\deg(F) < |G|$, we have*

$$\sum_{x \in G} F(x) = F(0) \cdot |G|.$$

The distinguisher in Proposition 4 can be regarded as an IPM integral distinguisher whose input set is $G \cup \{0\}$ and the mask is $\mathbf{c} = (1, 1, \dots, 1, 1, -|G|)$. In addition to forming integral distinguishers when $\deg(F) < |G|$, multiplicative subgroups may also play a role in scenarios where $\deg(F) \geq |G|$. Hence, we must describe the integral property of multiplicative subgroups more precisely.

Lemma 2. *Let p be a prime, x^i be a monomial in $\mathbb{F}_{p^n}[x]$ and G be a multiplicative subgroup of $\mathbb{F}_{p^n}$, $p^n > 2$, i is a non-negative integer, then the sum of x^i over G is*

$$\sum_{x \in G} x^i = \begin{cases} 0, & |G| \nmid i \\ |G|, & |G| \mid i \end{cases}$$

It should be noted that $\sum_{x \in G} x^0$ would not be 0 because the cardinality of G is not necessarily a multiple of p . This property is not desirable as, in most cases, the functions we analyze typically include a constant component. To address this issue, we can use an inner product mask.

Theorem 2. *Let p be a prime, G be a multiplicative subgroup of $\mathbb{F}_{p^n}$, consider the set $H = G \cup \{0\}$, then H belongs to an $((1, 1, \dots, 1, 1, -|G|), UD_{|G|}^{p^n} \cup \{0\})$ -IPM integral set with two levels.*

Generally, when set sizes are comparable, multiplicative subgroups exhibit better integral properties than additive subgroups, reflected by a larger indicator set E . Unlike additive subgroups with flexible size choices, multiplicative subgroups are constrained to sizes that are factors of the multiplicative group, limiting their selection. By analyzing the target block cipher and choosing suitable IPM integral sets, a more effective distinguisher can be achieved.

4.3 Creating Integral Sets with Better Integral Property

Having a wider range of options for integral sets will increase the probability of discovering superior integral distinguishers. In this section, we will introduce how to merge small sets to create a set with improved integral property.

Let $\mathbb{X}$ belong to an E -IPM integral set and $E_k^r : \mathbb{F}_{p^n} \rightarrow \mathbb{F}_{p^n}$ be the function of field-based block cipher. I represent the set of indices corresponding to non-zero coefficients. If $I - E$ is not empty, $\mathbb{X}$ will not form an integral distinguisher. However, if the set $I - E$ satisfies specific conditions, it is possible to construct an integral distinguisher by merging $\mathbb{X}$ and its cosets.

Proposition 5. *Let p be a prime, $\mathbb{X}$ be an E -IPM integral set whose elements belong to $\mathbb{F}_{p^n}$ and $a\mathbb{X} := \{ax : x \in \mathbb{X}\}$, $b\mathbb{X} := \{bx : x \in \mathbb{X}\}$ be its cosets, where $a, b \in \mathbb{F}_{p^n}$. Consider a new multiset $\mathbb{Y} = \mathbb{X} \cup a\mathbb{X} \cup b\mathbb{X}$, let Y be the set of elements in $\mathbb{Y}$ and $\mathbf{c}$ be a vector where its values denote the counts of each element in Y within $\mathbb{Y}$, if $a^i + b^i + 1 = 0$, then Y belongs to an $(\mathbf{c}, E \cup \{i, p \cdot i, p^2 \cdot i, \dots\})$ -IPM integral set.*

The proof of Proposition 5 is given in Appendix A.3. Proposition 5 shows how to construct a better set so that its indicator set can cover all the monomials of a certain function. However, merging also leads to larger cardinality, which means a higher complexity in attacks. Apart from merging, there may exist many other methods to handle sets for improved integral property. How to adjust a set to construct an integral distinguisher on certain cryptographic algorithms will be an interesting problem.

4.4 Combining with Monomial Detection Techniques

Determining the expression of a symmetric primitive is not an easy task. Fortunately, confirming the presence of certain monomials is sufficient to determine

an integral distinguisher. There have been several studies on monomial detection, such as division property, monomial prediction, and coefficient grouping. These techniques exhibit certain differences in terms of accuracy and efficiency. Given an E -IPM integral set and a block cipher $E_k^r(x)$, the algorithm to determine whether it forms an integral distinguisher over a certain number of rounds is given in Algorithm 1. The function *MonomialDetect()* in line 5 is responsible for monomial detection. When the monomial does not exist, it returns false; otherwise, it returns true. Cryptanalyst can choose any monomial detection technique they desire.

Algorithm 1 A framework for determining integral distinguishers

```

1: procedure ID-JUDGE(indicator set  $E$ , block cipher  $E_k^r(x)$ , the number of round
    $R$ )
2:    $flag \leftarrow 0$ 
3:    $\bar{E} \leftarrow [0, p^n - 1] \setminus E$ 
4:   for  $j \in \bar{E}$  do
5:      $flag \leftarrow MonomialDetect(Enc, R, j)$ 
6:     if  $flag == 1$  then
7:       return False
8:     end if
9:   end for
10:  return True
11: end procedure

```

5 Integral Attacks on MiMC, CIMINION and Chaghri

Previous integral distinguishers on ciphers over $\mathbb{F}_{2^n}$ are all based on additive subgroups. By applying IPM integral sets based on multiplicative subgroups and merged sets, we may find better distinguishers. In this section, we apply our framework to some famous block ciphers: MiMC, CIMINION and Chaghri. As a result, we successfully found some better integral distinguishers, exactly, the currently optimal integral distinguishers for the same number of rounds. A selection of our new distinguishers have been listed in Table 1 and the complete experimental results are given in Appendix F.

5.1 Application to MiMC

The authors of [1] give three types of MiMC in total, namely MiMC- n/n , MiMC- $2n/n$ (Feistel) and MiMC- p/p . We denote MiMC with exponent d and block size n as MiMC $_d(n, r)$ (or MiMC $_d(p, r)$ for MiMC over $\mathbb{F}_p$ respectively). The round function of MiMC $_d(n, r)$ can be written as

$$f_i(x_i) = (x_i \oplus k \oplus c_i)^d,$$

where c_i is the randomly chosen round constant. In the specification of MiMC, the designers allow exponents d of the form $2^l + 1$ and $2^l - 1$. The recommended value for block size is 129. According to the requirement of security, the number of rounds is $r = \lceil n \cdot \log_d 2 \rceil$.

Application to MiMC- n/n MiMC- n/n is the basic version of MiMC. In recent years, researchers have provided many different estimates of algebraic degree and have mounted integral attacks for MiMC- n/n [6, 8, 11]. A brief introduction of these works can be found in the Appendix C. Among them, Cui’s general monomial prediction (GMP) technique is the most powerful one.

We employ GMP as monomial detection method in Algorithm 1 and adjust the SMT models to search for IPM integral distinguishers based on multiplicative subgroups. By setting initial constraints as $u^{(0)} \% |G| = 0$ and $u^{(0)} > 0$, we can determine whether the set is an r -round integral distinguisher by solving only one SMT model. The detail of the searching algorithm is shown in Algorithm 2. The function $ModelGen(r, d)$ in line 5 would return the SMT model for $MiMC_d(129, r)$ ².

Algorithm 2 Explore the set of monomials $\{x^j : |G| \mid j, j > 0\}$ with GMP

```

1: procedure ID-GMP-MUL(The size of group  $|G|$ , block size  $n$ , the exponent of
   round function  $d$ )
2:    $R \leftarrow \lceil n \cdot \log_d 2 \rceil$ 
3:    $br \leftarrow \lfloor \log_d |G| \rfloor$ 
4:   while  $br \leq R$  do
5:      $M \leftarrow ModelGen(br, d)$ 
6:      $M.addConstr(u^{(0)} \% |G| = 0)$ 
7:      $M.addConstr(u^{(0)} > 0)$ 
8:     solve  $M$ 
9:     if  $M$  is satisfiable then
10:      return  $br - 1$ 
11:    end if
12:     $br \leftarrow br + 1$ 
13:  end while
14:  return  $R$ 
15: end procedure

```

We tried all different multiplicative subgroups and applied Algorithm 2 to $MiMC_3(129, r)$, the standard version of MiMC at first. The results are given in Table 3. To verify the results, we implemented the block cipher $MiMC_3(129, r)$ with SageMath [25] and tested the sum for a large number of randomly generated keys and round constants in small-scale instances. Across different keys

² As the process of generating the model is not the main focus, it will not be further discussed in this paper. Interested readers can refer to [8].

and round constants, we consistently obtained a sum of zero, which proves the correctness of our method.

We also applied Algorithm 2 to $\text{MiMC}_d(129, r)$ with different exponents $d \in \{5, 7, 9, 15, 17\}$, respectively. All the results are detailed in Table 4. According to Lucas's Theorem³, after one round of iteration, a power function with $d = 2^l - 1$ will generate more terms than the one with $d = 2^l + 1$. This implies that the output of x^{2^l-1} is more likely to become dense after iterating several rounds, which intuitively helps prevent IPM integral attacks. However, Table 4 shows that IPM integral sets based on multiplicative subgroups still prove effective when $d = 2^l - 1$. This indicates that the designer cannot resist our attack by simply replacing the round function of the block cipher.

When searching for integral distinguishers using a given multiplicative subgroup G , we found that some rounds cannot form an integral distinguisher due to the existence of a small number of monomials x^j satisfying $|G| \nmid j$. This issue can be resolved by merging sets according to Proposition 5, and the results obtained in this way are shown in Table 5. Table 5 shows that merging sets may yield better results than simply using larger multiplicative subgroups and additive subgroups in some cases. This once again emphasizes that exploiting input sets with various structures helps us obtain integral distinguishers with lower complexity. We also applied our framework on $\text{MiMC-}2n/n$ in a similar way and the details are given in Appendix F.3.

Application to $\text{MiMC-}p/p$ A variant of MiMC over prime finite fields is denoted as $\text{MiMC-}p/p$. MiMCHash over prime fields, a hash function based on MiMC, is one of the candidate hash functions in the STARKWARE hash challenge. To ensure the round function as a permutation, the prime number p should satisfy $\gcd(3, p-1) = 1$. When $\text{MiMC-}p/p$ is used to construct hash functions, the prime number p should also satisfy $p-1 = 2^l \cdot q$.

For a block cipher over $\mathbb{F}_p$, due to the property of $\mathbb{F}_p$, it is difficult for coefficients to cancel each other out. In other words, all monomials below the degree of block cipher will likely appear, which is completely different from the situation on $\mathbb{F}_2^n$. According to Proposition 4, we can construct distinguishers based on multiplicative subgroups if $\deg(F) < |G|$.

We can mathematically determine the lower bound of the complexity of IPM integral distinguishers for any r -round $\text{MiMC-}p/p$ when $3^r < p$. For an r -round $\text{MiMC-}p/p$, its degree is exactly 3^r and all monomials x^i , $i \leq 3^r$ will appear in its polynomial, which leads to strong resistance to integral attacks. According to Proposition 1, if we try to construct an IPM integral distinguisher, the infimum of its cardinality is $3^r + 2$ under our assumption.

³ By Lucas's Theorem, $\binom{n}{m} = \prod_{i=0}^k \binom{n_i}{m_i} \% 2$, where $n = \sum_{i=0}^k n_i \cdot 2^i$, $m = \sum_{i=0}^k m_i \cdot 2^i$ and $n_i, m_i \in \{0, 1\}$.

5.2 Application to CIMINION

CIMINION [10] is an MPC-friendly symmetric primitive based on Toffoli gates over large finite fields $\mathbb{F}_{2^n}$ and $\mathbb{F}_p$. The framework of CIMINION over $\mathbb{F}_{2^n}$ is described in Figure 1a. The construction is similar over $\mathbb{F}_p$ using modular addition. CIMINION takes a nonce $\mathcal{N}$ and two subkey elements K_1 and K_2 as input. It processes the input with permutation p_C to generate an intermediate state. Permutation p_E is then applied and truncated to encrypt plaintext elements P_1 and P_2 . To encrypt more elements, the intermediate state is updated by repeatedly adding two subkey elements and applying the rolling function rol and p_E to the resulting state. In this work, we only focus on permutations. Both p_C and p_E are the result of the repeated application of a round function. The only difference is the number of rounds. The number of rounds for p_C and p_E are $N = s + 6$ and $R = \max\{\lceil \frac{s+37}{12} \rceil, 6\}$, respectively, where s is the security level. We denote the function of round i as f_i (Figure 1b).

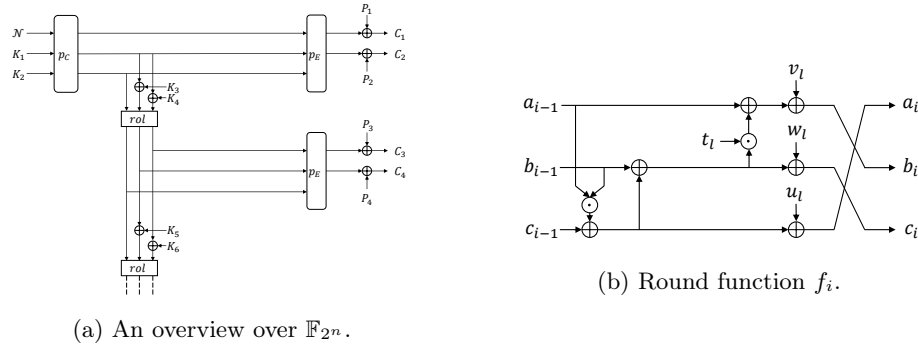


Fig. 1: CIMINION Encryption.

The attacker can choose the nonce $\mathcal{N}$ to construct an integral distinguisher. Therefore, nonce $\mathcal{N}$ is set as variable. Zhang et al. [30] derived an upper bound of $r - 1$ on the algebraic degree for the first two branches of r -th round p_C and constructed integral distinguishers through higher-order differentials, which is currently the best integral attack.

We conduct experiments on CIMINION over $\mathbb{F}_{2^{64}}$ and focus on the first branch. Firstly, we apply GMP to estimate the algebraic degree. The details of the experiment are given in Appendix D. Using GMP, we found that the previous estimated upper bounds on the algebraic degree of the first branch of p_C are slightly high in most cases.

Similarly to MiMC, we can also use IPM integral sets based on multiplicative subgroups and merged sets to construct more effective integral distinguishers for CIMINION. We adjust Algorithm 2 appropriately and apply it to CIMINION. As a consequence, we obtain many integral distinguishers with improved complexity,

and the results are given in Table 7. By applying IPM sets based on multiplicative subgroups and merged sets, we can obtain better results than those based on additive subgroups at different rounds. Compared to [30], we successfully achieved significant improvements on the complexity of distinguishers.

5.3 Application to Chaghri

Chaghri [3] is a FHE-friendly block cipher defined over a large finite field $\mathbb{F}_{2^{63}}$. We denote the state by $\mathbf{a} = (a_1, a_2, a_3) \in \mathbb{F}_{2^{63}}^3$. The number of rounds is 8 and each round is composed of two steps. The step function $S(\mathbf{a})$ of Chaghri⁴ is defined as $S(\mathbf{a}) = M \cdot B(G(\mathbf{a}))^T + \mathbf{RK}$, where $\mathbf{RK} = (RK_1, RK_2, RK_3) \in \mathbb{F}_{2^{63}}^3$ is the round key.

The components used in step function are the nonlinear function $G(\mathbf{a}) = (a_1^{2^{32}+1}, a_2^{2^{32}+1}, a_3^{2^{32}+1})$, the affine transform⁵ $B(\mathbf{a}) = (c_1 a_1^{2^3} + c_2, c_1 a_2^{2^3} + c_2, c_1 a_3^{2^3} + c_2)$ and a 3×3 MDS matrix M .

The coefficient grouping technique [20, 21], proposed and refined by Liu et al., is a novel method for monomial detection. It has been successfully employed to construct integral distinguishers targeting Chaghri. A brief introduction of the coefficient grouping technique is given in Appendix E.

To detect the existence of monomials with non-zero exponents that are divisible by the size of a certain multiplicative subgroup G , we need to add constraints $\mathcal{M}_n(\sum_{i=0}^{n-1} 2^i \gamma_i) \% |G| = 0$ and $\mathcal{M}_n(\sum_{i=0}^{n-1} 2^i \gamma_i) > 0$. However, the presence of $\mathcal{M}_n(\sum_{i=0}^{n-1} 2^i \gamma_i) \% |G| = 0$ prevents us from encoding the problem as an MILP problem as [20]. After carefully examining the operations and variable domains in the constraint problem, we have realized that SMT, supporting bit vectors and arithmetic operations, is a better choice. Therefore, we model the problem with CVC input language and solve it using STP as the solver. In particular, an algorithm for exploring the set of monomials $\{x^j : |G| \mid j, j > 0\}$ is shown in Algorithm 3.

Algorithm 3 includes two phases: constructing the basic model and finding all solutions. Lines 2-12 depict the process of constructing the SMT model with the limit of r -round coefficient $(N_{n-1}^r, N_{n-2}^r, \dots, N_0^r)$. On the other hand, lines 13-19 iteratively exclude feasible solutions until the model becomes unsatisfiable. The algorithm will output all monomials with non-zero exponent that are divisible by $|G|$. Based on the output, we can obtain IPM integral distinguishers using Theorem 2 and Proposition 5. The results on Chaghri are given in Table 8. With our new integral distinguishers, we can mount the integral attack on different steps of Chaghri with lower data and time complexities than [20].

⁴ Since the secure number of rounds depends on the security of decryption and low multiplicative depth in decryption desired in FHE schemes, we consider the decryption of Chaghri as [20].

⁵ $c_1, c_2 \in \mathbb{F}_{2^{63}}$ are constants. After the proposal of coefficient grouping [20], the affine transform has been replaced by a denser one. However, we focus on the original version to make a better comparison with the result in [20].

Algorithm 3 Explore the set of monomials $\{x^j : |G| \mid j, j > 0\}$ with coefficient grouping

```

1: procedure ID-CP-MUL(The size of group  $|G|$ , the  $r$ -th round coefficient limit
   ( $N_{n-1}^r, N_{n-2}^r, \dots, N_0^r$ ))
2:    $M \leftarrow$  an empty model
3:    $R \leftarrow \emptyset$ 
4:   for  $i$  from 0 to  $n - 1$  do
5:      $M.addVar(a_i)$ 
6:      $M.addConstr(a_i \geq 0)$ 
7:      $M.addConstr(a_i \leq N_i^r)$ 
8:   end for
9:    $M.addVar(s)$ 
10:   $M.addConstr(s = \sum_{i=0}^{n-1} 2^i a_i)$ 
11:   $M.addConstr(s > 0)$ 
12:   $M.addConstr(s \% |G| = 0)$ 
13:  solve  $M$ 
14:  while  $M$  is satisfiable do
15:     $t \leftarrow M.getVal(s)$ 
16:     $R \leftarrow R \cup \{t\}$ 
17:     $M.addConstr(s \neq t)$ 
18:    solve  $M$ 
19:  end while
20:  return  $R$ 
21: end procedure

```

6 Conclusion

Considering the research effort in this domain for the past two decades, when the integral attacks are referred to, mostly we consider the higher-order differential attacks. However, recent results show that additive subgroups are no longer the best choice for integral attacks on MPC/FHE/ZK-friendly symmetric primitives. It becomes more and more evident that exploiting different structures as inputs can lead to improved cryptanalytic results. This is the context of this work. In this work, we propose the concept of IPM integral distinguishers and analysis the existence of such distinguishers under different coefficient matrices. In order to construct IPM integral distinguishers for practical block ciphers, we introduce the concept of IPM integral sets and a framework for seeking integral distinguishers. Applying our framework to some AO ciphers: MiMC, CIMINION, and Chaghri, we achieve several distinguishers with lower complexity by using IPM integral sets based on multiplicative subgroups and merged sets. It should be noted that our framework can perfectly adapt to various monomial detection techniques. From this fact, we expect that the IPM integral sets will be of great help in improving integral attacks. There are indeed several open questions that require further research. For example, whether there are other classes of IPM integral sets and how to construct E -IPM integral sets for any indicator set E . These we leave for future research.

Acknowledgments

The authors would like to thank the anonymous reviewers for their valuable suggestions and comments. The work of Deng Tang was supported in part by the National Key R&D Program of China (No. 2020YFA0712300) and NSFC (Nos. 62272303, 12031011).

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A Proofs

A.1 Proof of Proposition 1

Proof. Suppose the input set of plaintexts is $\mathbb{X} = \{x_1, x_2, \dots, x_t\}$ and $x_i \neq x_j, \forall i \neq j$, we have

$$M = \begin{pmatrix} x_1^0 & x_2^0 & \cdots & x_t^0 \\ x_1^1 & x_2^1 & \cdots & x_t^1 \\ \vdots & \vdots & & \vdots \\ x_1^d & x_2^d & \cdots & x_t^d \end{pmatrix} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_t \\ \vdots & \vdots & & \vdots \\ x_1^d & x_2^d & \cdots & x_t^d \end{pmatrix}.$$

M is a Vandermonde matrix of order $d+1$ -by- t .

When $t \leq d+1$, the rank of matrix $R(M) = \min\{d+1, t\} = t$. Therefore, the equation $M\mathbf{c} = \mathbf{0}$ only have a zero solution.

When $t > d+1$, the rank of matrix $R(M) = \min\{d+1, t\} = d+1 < t$. The equation $M\mathbf{c} = \mathbf{0}$ would have more than one solutions. Hence, it is possible to construct an IPM integral distinguisher, and constructing an IPM integral distinguisher requires at least $d+2$ elements.

A.2 Proof of Proposition 2

Proof. Suppose the input set of plaintexts is $\mathbb{X} = \{x_1, x_2, \dots, x_t\}$ and $x_i \neq x_j, \forall i \neq j$, we have

$$M = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1^{i_1} & x_2^{i_1} & \cdots & x_t^{i_1} \\ \vdots & \vdots & & \vdots \\ x_1^{i_{|I|}} & x_2^{i_{|I|}} & \cdots & x_t^{i_{|I|}} \end{pmatrix}.$$

Denote the contiguous sequences with maximum length of I is $(i_j, i_j + 1, i_j + 2, \dots, i_j + m - 1)$. By choosing these rows, we get another submatrix M'

$$M' = \begin{pmatrix} x_1^{i_j} & x_2^{i_j} & \cdots & x_t^{i_j} \\ x_1^{i_j+1} & x_2^{i_j+1} & \cdots & x_t^{i_j+1} \\ \vdots & \vdots & & \vdots \\ x_1^{i_j+m-1} & x_2^{i_j+m-1} & \cdots & x_t^{i_j+m-1} \end{pmatrix}.$$

It is clear that $R(M) \geq R(M')$ and $\det(M') = x_1^{i_j} x_2^{i_j} \cdots x_t^{i_j} \prod_{1 \leq u < v \leq t} (x_v - x_u)$. Obviously, whether the zero element is in $\mathbb{X}$ will determine whether $\det(M') = 0$, so we need to discuss these two cases separately.

Firstly, if $0 \notin \mathbb{X}$, then $\det(M') \neq 0$. When $t \leq m$, $R(M') = \min\{m, t\} = t$. Therefore, we have $R(M) = t$, and the equation $M\mathbf{c} = \mathbf{0}$ only have a zero solution. Only when $t \geq m + 1$, is it possible for equation $M\mathbf{c} = \mathbf{0}$ to have a non-zero solution.

Secondly, if $0 \in \mathbb{X}$, then $\det(M') = 0$. Suppose $x_1 = 0$, we have $M' = \mathbf{0} || M''$, where $||$ denotes concatenation, M'' is a m -by- $(t-1)$ matrix. The determinant of M'' is computed as $\det(M'') = x_2^{i_j} x_3^{i_j} \cdots x_t^{i_j} \prod_{2 \leq u < v \leq t} (x_v - x_u) \neq 0$. Similarly, we know that only when $t \geq m + 2$, the equation $M\mathbf{c} = \mathbf{0}$ will have a non-zero solution.

In conclusion, it requires at least $m + 1$ elements to construct an IPM integral distinguisher.

A.3 Proof of Proposition 5

Proof. Because $\mathbb{Y} = \mathbb{X} \cup a\mathbb{X} \cup b\mathbb{X}$, we have

$$\sum_{x \in \mathbb{Y}} x^j = \sum_{x \in \mathbb{X}} x^j + \sum_{x \in a\mathbb{X}} x^j + \sum_{x \in b\mathbb{X}} x^j = \sum_{x \in \mathbb{X}} (x^j + (ax)^j + (bx)^j) = (1 + a^j + b^j) \sum_{x \in \mathbb{X}} x^j.$$

For $j \in E$, it is clear that

$$\sum_{x \in \mathbb{Y}} x^j = (1 + a^j + b^j) \sum_{x \in \mathbb{X}} x^j = (1 + a^j + b^j) \cdot 0 = 0.$$

For $j \in \{i, p \cdot i, p^2 \cdot i, \dots\}$, there exists l such that $j = p^l \cdot i$ and the sum becomes

$$\sum_{x \in \mathbb{Y}} x^j = (1 + a^j + b^j) \sum_{x \in \mathbb{X}} x^j = (1 + a^{p^l \cdot i} + b^{p^l \cdot i}) \sum_{x \in \mathbb{X}} x^j = (1 + a^i + b^i)^{p^l} \sum_{x \in \mathbb{X}} x^j = 0.$$

Equation $(1 + a + b)^p = 1 + a^p + b^p$ holds as the characteristic of $\mathbb{F}_{p^n}$ is p . Hence, for $j \in E \cup \{i, p \cdot i, p^2 \cdot i, \dots\}$, $\sum_{i=1}^{|Y|} c_i x_i^j = 0$.

B A Comparison Between IPM Integral Sets and Division Property

At first, we give the definition of IPM integral set with two levels.

Definition 4 (IPM Integral Set with Two Levels). Given a set $\mathbb{X} = \{x_1, x_2, \dots, x_t\} \subseteq \mathbb{F}_{p^n}$, a set $E \subseteq [0, p^n - 1]$ and a vector $\mathbf{c} \in \mathbb{F}_{p^n}^t$, if the set $\mathbb{X}$ satisfies:

$$\sum_{j=1}^t c_j x_j^i = \begin{cases} 0, & i \in E \\ c, & \text{otherwise} \end{cases}$$

where $c \in \mathbb{F}_{p^n}$ is a constant, we call $\mathbb{X}$ belongs to an $(\mathbf{c}, E)$ -IPM integral set with two levels, or $\mathbb{X}$ is an $(\mathbf{c}, E)$ -IPM integral set with two levels.

An example for IPM integral set with two levels is also described in Appendix G.3. The values in the non-zero part don't need to be fixed constants when directly constructing an integral distinguisher while combining different two-level IPM integral sets is more likely to yield a superior integral set. Some common versions of the division property are listed as below.

Definition 5 (Word-based Division Property). Let $\mathbb{X}$ be a multiset whose elements take a value of $\mathbb{F}_2^m$, and k takes a value between 0 and n . When the multiset $\mathbb{X}$ has the word-based division property $\mathcal{D}_k^m$, it fulfils the following conditions:

$$\bigoplus_{x \in \mathbb{X}} \mathbf{x}^{\mathbf{u}} = \begin{cases} \text{unknown}, & \text{if } wt(\mathbf{u}) \geq k \\ 0, & \text{otherwise.} \end{cases}$$

Definition 6 (Bit-based Division Property). Let $\mathbb{X}$ be a multiset whose elements take a value of $\mathbb{F}_2^m$, and $\mathbb{K} \subseteq \mathbb{F}_2^m$. When the multiset $\mathbb{X}$ has the bit-based division property $\mathcal{D}_{\mathbb{K}}^m$, it fulfils the following conditions:

$$\bigoplus_{x \in \mathbb{X}} \mathbf{x}^{\mathbf{u}} = \begin{cases} \text{unknown}, & \text{if there are } \mathbf{k} \in \mathbb{K} \text{ s.t. } \mathbf{u} \succeq \mathbf{k} \\ 0, & \text{otherwise.} \end{cases}$$

Definition 7 (Three-subset Division Property Without Unknown Set). Let $\mathbb{X}$ be a multiset whose elements take a value of $\mathbb{F}_2^m$. Let $\tilde{\mathbb{L}}$ be also a multiset whose elements take a value of $\mathbb{F}_2^m$. When the multisets $\mathbb{X}$ has the three-subset division property without unknown set $\mathcal{T}_{\tilde{\mathbb{L}}}^{1^m}$, it fulfils the following conditions

$$\bigoplus_{x \in \mathbb{X}} \mathbf{x}^{\mathbf{u}} = \begin{cases} 1, & \text{if there are odd-number of } \mathbf{u}'\text{'s in } \tilde{\mathbb{L}} \\ 0, & \text{otherwise.} \end{cases}$$

Compared to the word-based division property (as defined in Definition 5), the IPM integral set is more refined. Firstly, the word-based division property treats the non-zero sum value as “unknown”, while the IPM integral set can calculate the exact value of the sum. Secondly, the word-based division property partitions the exponents solely based on Hamming weights, whereas the IPM integral set partitions them using an indicator set E . In summary, the IPM integral set can provide a more accurate characterization of integral property than the word-based division property.

To compare it with the bit-based division property (as defined in Definitions 6, 7), we firstly define *bit-based IPM integral set*.

Definition 8 (Bit-based IPM Integral Set). *Given a set $\mathbb{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_t\} \subseteq \mathbb{F}_2^n$, a set $E \subseteq \{0, 1\}^n$, and a matrix $c \in \mathbb{F}_2^t$, if the set $\mathbb{X}$ satisfies $\bigoplus_{i=1}^t c_i \mathbf{x}_i^{\mathbf{j}} = 0$ for all $\mathbf{j} \in E$ and $\bigoplus_{\mathbf{x} \in \mathbb{X}} \mathbf{x}^{\mathbf{j}}$ is always 1 for $\mathbf{j} \in \{0, 1\}^n \setminus E$, then we define that $\mathbb{X}$ belongs to an $(\mathbf{c}, E)$ -bit-based IPM integral set, or $\mathbb{X}$ is an $(\mathbf{c}, E)$ -bit-based IPM integral set.*

Since the only non-zero element in $\mathbb{F}_2$ is 1, there is no difference between the bit-based IPM integral set and the bit-based IPM integral set with two levels. According to Definition 8, it is evident that when $\mathbf{c} = \mathbf{1}$, the bit-based IPM integral set is essentially equivalent to the three-subset division property without unknown set, where the complement of E corresponds to the set of elements that appear an odd number of times in $\mathbb{L}$ in the latter case. Therefore, the core idea behind the IPM integral set can be regarded as a generalization of three-subset division property without unknown set over $\mathbb{F}_{p^n}$.

C Previous Estimates of Algebraic Degree for MiMC

In [11], Eichlseder et al. stated that the algebraic degree of MiMC grows linearly in the number of rounds, and thus the security margin against is only 1 or 2 rounds. Let $\delta_{[EGL+20]}^{(d,r)}$ denote the theoretical upper bound of algebraic degree in [11]. Then the bound is computed as $\delta_{[EGL+20]}^{(d,r)} = \lfloor \log_2(d^r + 1) \rfloor$.

Besides the bound $\delta_{[EGL+20]}^{(d,r)}$, Bouvier et al. [6] proposed another theoretical bound of algebraic degree for the case $d = 2^l + 1$, denoted by $\delta_{[BCP23]}^{(d,r)} \cdot \delta_{[BCP23]}^{(d,r)}$ is tighter than $\delta_{[EGL+20]}^{(d,r)}$ and computed as

$$\delta_{[BCP23]}^{(d,r)} = \begin{cases} 2 \cdot \lceil k_r/2 - 1 \rceil, & l = 1, \\ \lfloor r \log_2 d \rfloor - l + 1, & l \geq 2, \end{cases}$$

where $k_r = \lfloor r \log_2 d \rfloor$.

The SOTA work about integral attacks on MiMC is based on field-based division property [8]. Cui et al. proposed a method for finding integral distinguishers over $\mathbb{F}_{2^n}$, called general monomial prediction (GMP). They decomposed the block cipher into a sequence of simple functions and proposed the corresponding propagation rules of monomials. With the aid of the bit-vector theory of Satisfiability Modulo Theories (SMT), they successfully modeled the propagation of monomials using CVC [4] input language and took STP [7, 12] and Cryptominisat5 [24] as solvers. The framework of Cui's degree estimation method is given in Algorithm 4.

Particularly, an initial constraint $wt(u^{(0)}) = l$ needs to be added into the SMT model for detecting the monomials with l Hamming weight, where $u^{(0)}$ is the exponent of input variable $x^{(0)}$. The upper bound found by GMP denoted as $\delta_{GMP}^{(d,r)}$, is the best upper bound so far. Moreover, $\delta_{GMP}^{(d,r)}$ exactly equals the observed algebraic degree in small-scale instances [8], which highlights the accuracy of GMP.

Algorithm 4 Degree estimation method in [8]

```

1: procedure DEG-EST(The  $r$ -round SMT model  $M_r$ , the maximum algebraic degree
    $\Delta$ )
2:    $i \leftarrow \lfloor \log_2(d^r + 1) \rfloor$ 
3:   while  $i > 0$  do
4:      $M \leftarrow M_r$ 
5:      $M.addConstr(wt(u^{(0)}) = i)$ 
6:     solve  $M$ 
7:     if  $M$  is satisfiable then
8:       return  $i$ 
9:     end if
10:     $i \leftarrow i - 1$ 
11:  end while
12:  return 0
13: end procedure
    
```

D Applying GMP on CIMINION

For each $i \geq 1$, the round function $f_i : \mathbb{F}_q^3 \rightarrow \mathbb{F}_q^3$ of CIMINION can be written as:

$$\begin{bmatrix} a_i \\ b_i \\ c_i \end{bmatrix} := \begin{bmatrix} 0 & 0 & 1 \\ 1 & t_l & t_l \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} a_{i-1} \\ b_{i-1} \\ c_{i-1} + a_{i-1} \cdot b_{i-1} \end{bmatrix} + \begin{bmatrix} u_l \\ v_l \\ w_l \end{bmatrix}, \quad (2)$$

where t_l, u_l, v_l, w_l are round constants and $l = i$ for p_C , $l = i + N - R$ for p_E . For simplicity, we will take $l = i$ in the following content.

We would use $\pi_{(x^{(i)}, y^{(i)}, z^{(i)})}(a_i, b_i, c_i)$ to denote the monomials of i -th round state and $\pi_{p^{(i)}}(t_i), \pi_{q^{(i)}}(u_i), \pi_{r^{(i)}}(v_i), \pi_{s^{(i)}}(w_i)$ to denote the monomial of i -th round constant. With some auxiliary variables, the SMT model M_r is described as Equation (3).

$$\mathcal{M}_r \leftarrow \begin{cases} (x^{(i)}) \xrightarrow{COPY} (x_0^{(i)}, x_1^{(i)}) & \text{for } 0 \leq i < r, \\ (y^{(i)}) \xrightarrow{COPY} (y_0^{(i)}, y_1^{(i)}) & \text{for } 0 \leq i < r, \\ (x_0^{(i)}, y_0^{(i)}) \xrightarrow{MUL} (d^{(i)}) & \text{for } 0 \leq i < r, \\ (z^{(i)}, d^{(i)}) \xrightarrow{XOR} (e^{(i)}) & \text{for } 0 \leq i < r, \\ (e^{(i)}) \xrightarrow{COPY} (e_0^{(i)}, e_1^{(i)}) & \text{for } 0 \leq i < r, \\ (e_0^{(i)}, y_1^{(i)}) \xrightarrow{XOR} (f^{(i)}) & \text{for } 0 \leq i < r, \\ (f^{(i)}) \xrightarrow{COPY} (f_0^{(i)}, f_1^{(i)}) & \text{for } 0 \leq i < r, \\ (f_0^{(i)}, p^{(i)}) \xrightarrow{MUL} (g^{(i)}) & \text{for } 0 \leq i < r, \\ (g^{(i)}, x_1^{(i)}) \xrightarrow{XOR} (h^{(i)}) & \text{for } 0 \leq i < r, \\ (e_1^{(i)}, q^{(i)}) \xrightarrow{XOR} (x^{(i+1)}) & \text{for } 0 \leq i < r, \\ (h^{(i)}, r^{(i)}) \xrightarrow{XOR} (y^{(i+1)}) & \text{for } 0 \leq i < r, \\ (f_1^{(i)}, s^{(i)}) \xrightarrow{XOR} (z^{(i+1)}) & \text{for } 0 \leq i < r, \end{cases} \quad (3)$$

Compared with the SMT models for MiMC and Feistel MiMC, the one for CIMINION is more complex. For one round CIMINION, there are a total of 12 basic operations and 20 variables. The increase in the number of variables leads to a significantly larger solution space, which directly results in a slower solving process. Particularly when $r \geq 55$, the Algorithm 4 is quite time-consuming. Therefore, how to efficiently build SMT models is a worthwhile research topic.

By setting the initial constraints and stopping rules properly, we use GMP to search the algebraic degree at first. As a result, we derive some new upper bounds on the algebraic degree of the first branch of p_C and summarize them in Table 2.

Table 2: Upper bounds on algebraic degree of the first branch of p_C

Number of rounds	1	2	3	4	5	6	$\dots$	i	$\dots$	r	$r+1$
Previous result [30]	1	1	2	3	4	5	$\dots$	$i-1$	$\dots$	$r-1$	r
Our result	1	1	2	2	3	4	$\dots$	$i-2$	$\dots$	$r-2$	$r-1$

As shown in Table 2, the new upper bound on the algebraic degree we obtained grows linearly with the number of rounds, but it is 1 less than the previous result starting from the 4-th round. The linear growth can be attributed to the fact that the round function is a quadratic function, which is in line with the designer's statement [10]. Next, we will focus on analyzing why the algebraic degree of a_4 is 2 rather than 3.

By setting a_0 as variable x , the univariate degree of the first branch a_4 is 8, and only when the monomial x^7 appears will the algebraic degree be 3. From Equation (2), we know that $a_i = u_i \oplus c_{i-1} \oplus a_{i-1}b_{i-1}$. Since the univariate degree of a_3, b_3, c_3 are all 4, the monomial x^7 can only be generated from the term a_3b_3 . More specifically, it can only be generated from the multiplication of x^3 and x^4 in a_3, b_3 . By iteratively computed, we obtain that the coefficients of x^3 and x^4 in a_3 are

$$p_1 = b_0^3 t_1^2 t_2 \oplus b_0^3 t_1 \oplus b_0^2 \oplus b_0 t_2, p_2 = b_0^4 t_1^2 t_2 \oplus b_0^2 t_2,$$

and the coefficients of x^3 and x^4 in b_3 are

$$q_1 = b_0^3 t_1^2 t_2 t_3 \oplus b_0^3 t_1 t_3 \oplus b_0^2 t_3 \oplus b_0 t_2 t_3, q_2 = b_0^4 t_1^2 t_2 t_3 \oplus b_0^2 t_2 t_3.$$

By observing the coefficients, we have $q_1 = t_3 p_1, q_2 = t_3 p_2$. Therefore, the coefficient of x^7 in a_4 can be calculated as

$$p_1 q_2 \oplus p_2 q_1 = t_3 p_1 p_2 \oplus t_3 p_1 p_2 = 0,$$

which means x^7 will never appear in a_4 . Hence, the algebraic degree a_4 is at most 2. Because the only nonlinear term for all branches in the round function is $a_{i-1}b_{i-1}$, similar results can be obtained for the other two branches.

E The Coefficient Grouping Technique

The main idea of coefficient grouping is to give an expression of all possible monomials and find the maximum Hamming weight among the exponents. To be specific, by setting the input state as

$$a_1^{(0)} = A_{0,1}X + B_{0,1}, a_2^{(0)} = A_{0,2}X + B_{0,2}, a_3^{(0)} = A_{0,3}X + B_{0,3},$$

where $X \in \mathbb{F}_{2^{63}}$ is the variable and $A_{0,i}, B_{0,i} \in \mathbb{F}_{2^{63}} (1 \leq i \leq 3)$ are randomly chosen constants, the set of possible exponents in r -th round is

$$w_r = \{e : e = \mathcal{M}_n(\sum_{i=0}^{n-1} 2^i \gamma_i), 0 \leq \gamma_i \leq N_i^r\}. \quad (4)$$

$\mathcal{M}_n(x)$ is a function defined as follows:

$$\mathcal{M}_n(x) = \begin{cases} 2^n - 1, & (2^n - 1) \nmid x \text{ and } x \neq 0, \\ x \% (2^n - 1), & \text{otherwise.} \end{cases}$$

In Equation (4), the vector of integers $(N_{n-1}^r, N_{n-2}^r, \dots, N_0^r)$ can be computed by the recursive relation

$$N_i^{j+1} = N_{(i-35)\%n}^j + N_{(i-32)\%n}^j, 0 \leq i \leq n-1, j \geq 0$$

and the initial value is

$$N_0^0 = 1, N_i^0 = 0, 1 \leq i \leq n-1.$$

With the value of $(N_{n-1}^r, N_{n-2}^r, \dots, N_0^r)$, the algebraic degree can be obtained by finding the maximum Hamming weight. In [20], the authors encoded the problem as a Mixed Integer Linear Programming (MILP) problem and presented a method for modeling arithmetic addition and comparison as MILP constraints.

F Experimental Results

F.1 Results on MiMC₃(129, r)

The result on MiMC₃(129, r) is listed in Table 3. In the subsequent results presentation, we will use **bold font** to highlight significant improvements.

F.2 Results on MiMC _{d} (129, r) for different d

F.3 Results on FeistelMiMC₃(129, r)

The designers of MiMC also proposed MiMC-2 n/n , or Feistel MiMC, whose non-linear permutation is the same as MiMC. We denote Feistel MiMC with exponent d and block size n as FeistelMiMC _{d} (n , r). The i -th round function of FeistelMiMC _{d} (n , r) is defined as

$$(x_0^{(i+1)}, x_1^{(i+1)}) \leftarrow (x_1^{(i)} \oplus (x_0^{(i)} \oplus k \oplus c_i)^d, x_0^{(i)}).$$

Table 3: Effective integral distinguishers found on $\text{MiMC}_3(129, r)$.

Round	$\delta_{GMP}^{(d,r)}$	Comp. of HD-ID*	$ G $ *	Comp. of IPM-ID†
3	4	2^5	p_1	2^3
6	8	2^9	p_2	$2^{8.75}$
16	24	2^{25}	$p_1 \cdot p_2 \cdot p_3$	$2^{24.81}$
21	32	2^{33}	$p_1 \cdot p_2 \cdot p_4$	$2^{32.56}$
60	94	2^{95}	$p_1 \cdot p_2 \cdot p_5$	$2^{94.75}$

* The complexity of integral distinguishers based on additive subgroups is computed as $2^{\delta_{GMP}^{(d,r)}+1}$, where $\delta_{GMP}^{(d,r)}$ is the estimated upper bound of algebraic degree.

* $(p_1, p_2, p_3, p_4, p_5) = (7, 431, 9719, 2099863, 11053036065049294753459639)$ are factors of $2^{129} - 1$.

† According to Theorem 2, a zero element needs to be added. Hence, the complexity of integral distinguishers based on multiplicative subgroups is computed as $2^{\lceil \log_2(|G|+1) \rceil}$.

The round constants c_i are again chosen randomly except for the first and last round constants which are equal to 0. The number of rounds for $\text{FeistelMiMC}_d(n, r)$ is $r' = 2 \cdot r = 2 \cdot \lceil n \cdot \log_d 2 \rceil$.

We focus on the univariate case of Feistel MiMC. In this case, we only consider the input and output of the left branch. Due to the structure of the Feistel network, all integral distinguishers found in this way can be naturally extended with 2 more rounds. Without a doubt, when considering the same nonlinear function and number of rounds, the output of the left branch of Feistel MiMC is denser than that of MiMC. New variable x_1 and the addition of both branches decrease the probability of two identical monomials canceling each other out. Consider the case where $r = 3$. The set of monomials $\{x^5, x^7, x^{13}\}$ will appear in $\text{FeistelMiMC}_3(n, r)$, but not in $\text{MiMC}_3(n, r)$. However, this characteristic does not prevent us from finding better results. We apply Algorithm 2 to $\text{FeistelMiMC}_3(n, r)$, and the results are shown in Table 6.

For Feistel MiMC, we can also obtain distinguishers with lower complexity than those of [8]. This indicates that our new framework is a universal framework that can optimize integral attacks on ciphers with different structures.

F.4 Results on CIMINION

F.5 Results on Chaghri

We denote the algebraic degree upper bound of r -step Chaghri as δ_r . Applying Algorithm 3 to Chaghri, we obtain some IPM integral distinguishers which are more effective than previous ones in [20]. The results are listed in Table 8.

Table 4: Effective integral distinguishers found on different versions of $\text{MiMC}_d(129, r)$.

d	Round	$\delta_{GMP}^{(d,r)}$	Comp. of HD-ID	$ G $	Comp. of IPM-ID
	2	3	2^4	$\mathbf{p}_1$	2^3
5	41	94	2^{95}	$p_1 \cdot p_2 \cdot p_5$	$2^{94.75}$
	49	112	2^{113}	$p_2 \cdot p_4 \cdot p_5$	$2^{112.95}$
7	9	24	2^{25}	$p_1 \cdot p_2 \cdot p_3$	$2^{24.81}$
	17	45	2^{46}	$p_1 \cdot p_2 \cdot p_3 \cdot p_4$	$2^{45.81}$
9	2	3	2^4	$\mathbf{p}_1$	2^3
	11	32	2^{33}	$p_1 \cdot p_2 \cdot p_4$	$2^{32.56}$
15	29	112	2^{113}	$p_2 \cdot p_4 \cdot p_5$	$2^{112.95}$
	31	120	2^{121}	$p_1 \cdot p_3 \cdot p_4 \cdot p_5$	$2^{120.25}$
17	2	3	2^4	$\mathbf{p}_1$	2^3
	12	44	2^{45}	$\mathbf{p}_2 \cdot \mathbf{p}_3 \cdot \mathbf{p}_4$	2^{43}

G Examples

G.1 An Example for IPM Integral Distinguishers

Example 1. Let $f(x)$ be a polynomial over $\mathbb{F}_{2^4}[x]$ with the irreducible polynomial $p(x) = x^4 + x + 1$ and $f(x) = a_0 + a_1x^3 + a_2x^5 + a_3x^7$, where $\{a_i\}_{i=0,1,2,3}$ are arbitrary constants over $\mathbb{F}_{2^4}$ and the primitive element of $\mathbb{F}_{2^4}$ is α . If we use the mask $\mathbf{c} = \{\alpha, \alpha^3, \alpha^{10}, \alpha^{10}, 1\}$ and the set $\mathbb{X} = \{\alpha, \alpha^2, \alpha^3, \alpha^4, \alpha^5\}$, we can get an IPM integral distinguisher with only 5 elements.

Because the algebraic degree of $f(x)$ is 3, 2^4 elements are needed if we use additive subgroups to form an integral distinguisher, and multiplicative subgroups would not perform better since the existence of x^3 and x^5 . A more comprehensive description of the integral property of additive and multiplicative subgroups is presented in Section 4.2. With an inner product mask, it only needs 5 elements to form an integral distinguisher for $f(x)$, greatly decreasing the complexity. Therefore, Example 1 demonstrates the power of inner product masks.

G.2 An Example for IPM Integral Distinguisher with $|\mathbb{X}| \leq |I|$

Example 2. Let $f(x)$ be a polynomial over $\mathbb{F}_{2^9}[x]$ with the irreducible polynomial $p(x) = x^9 + x^4 + 1$ and $f(x) = a_0 + a_1x + a_2x^3 + a_3x^7 + a_4x^{16} + a_5x^{34} + a_6x^{61} + a_7x^{73} + a_8x^{90} + a_9x^{127} + a_{10}x^{141} + a_{11}x^{170} + a_{12}x^{185} + a_{13}x^{208} + a_{14}x^{255} + a_{15}x^{284} + a_{16}x^{317} + a_{17}x^{351} + a_{18}x^{386} + a_{19}x^{412} + a_{20}x^{414} + a_{21}x^{448} + a_{22}x^{471} + a_{23}x^{509}$, where $\{a_i\}_{i=0,1,\dots,23}$ are arbitrary constants over $\mathbb{F}_{2^9}$ and α is the primitive element of $\mathbb{F}_{2^9}$. We can form an IPM integral distinguisher with a set $X = \{\alpha, \alpha^2, \alpha^4, \alpha^8, \alpha^{16}, \alpha^{32}, \alpha^{64}, \alpha^{128}, \alpha^{256}, 0\}$ and a mask $\mathbf{c} = \{1, 1, 1, 1, 1, 1, 1, 1, 1\}$.

Table 5: Effective integral distinguishers found on different versions of $\text{MiMC}_d(129, r)$ by merging sets.

d	Round	$\delta_{GMP}^{(d,r)}$	Comp. of HD-ID	$ G $	Monomials*	Comp. of IPM-ID*
3	7	10	2^{11}	p_2	$\{x^{2 G }, x^{4 G }\}$	$2^{10.33}$
	64	100	2^{101}	$p_1 \cdot p_3 \cdot p_5$		$2^{100.83}$
5	20	44	2^{45}	$p_2 \cdot p_3 \cdot p_4$	$\{x^{4 G }, x^{8 G }\}$	$2^{44.59}$
15	1	4	2^5	p_1	$\{x^{ G }, x^{2 G }\}$	$2^{4.46}$
	12	44	2^{45}	$p_2 \cdot p_3 \cdot p_4$	$\{x^{2 G }, x^{4 G }, x^{8 G }\}$	$2^{44.59}$

* The set of monomials $\{x^j : |G| \mid j\}$ that exists in $\text{MiMC}_d(129, r)$.

* According to Proposition 5, the size of new merged multiset is $3 \cdot (|G| + 1)$. With an inner product mask, this value can be reduced to $3|G| + 1$.

Table 6: Effective integral distinguishers found on $\text{FeistelMiMC}_3(129, r)$.

Round	$\delta_{x_0}(x_0^{(r)})^*$	Comp. of HD-ID	$ G $	Comp. of IPM-ID
7	11	2^{12}	$p_1 \cdot p_2$	$2^{11.56}$
15	23	2^{24}	$p_1 \cdot p_4$	$2^{23.81}$
58	91	2^{92}	$p_2 \cdot p_5$	$2^{91.94}$
71	112	2^{113}	$p_2 \cdot p_4 \cdot p_5$	$2^{112.95}$
73	115	2^{116}	$p_1 \cdot p_2 \cdot p_4 \cdot p_5$	$2^{115.75}$
74	117	2^{118}	$p_3 \cdot p_4 \cdot p_5$	$2^{117.74}$

* We denote the algebraic degree of left branch over $x_0^{(0)}$ as $\delta_{x_0}(x_0^{(r)})$.

G.3 Examples for IPM Integral Sets

Example 3. Let α be a primitive element of $\mathbb{F}_{2^4}$ with the irreducible polynomial $f(x) = x^4 + x + 1$. Let $\mathbb{X} \subseteq \mathbb{F}_{2^4}$. As an example, we prepare the set as

$$\mathbb{X} := \{0, \alpha, \alpha^6, \alpha^{11}\}.$$

Table 9 calculates the sum of x^j for different j .

For all $j \in H = UD_3^{15} \cup \{0\}$, the sum $\bigoplus_{x \in \mathbb{X}} x^j$ becomes 0. Therefore, $\mathbb{X}$ is an H -IPM integral set.

For an IPM integral set with two levels, consider the following example: $\mathbb{Y} := \{0, 1, \alpha^5, \alpha^{10}\}$, where α is a primitive element of $\mathbb{F}_{2^4}$ with the irreducible polynomial $f(x) = x^4 + x + 1$. In the case of $\mathbb{Y}$, its indicator set is the same as $\mathbb{X}$ in Example 3, while $\bigoplus_{y \in \mathbb{Y}} y^j$ always evaluates to 1 in the non-zero part.

Table 7: Effective integral distinguishers found on CIMINION over $\mathbb{F}_{2^{64}}$

Round	Results in [30]	Comp. of HD-ID *	$ G $ *	Comp. of IPM-ID
3	2^3	2^3	q_1	$2^{2.58}$
23	2^{23}	2^{22}	$q_3 \cdot q_4 \cdot q_5$	$2^{21.42}$
39	2^{39}	2^{38}	$q_3 \cdot q_4 \cdot q_5 \cdot q_6$	$2^{37.42}$
51	2^{51}	2^{50}	$q_1 \cdot q_5 \cdot q_6 \cdot q_7$	$2^{49.58}$
63	2^{63}	2^{62}	$q_1 \cdot q_3 \cdot q_4 \cdot q_5 \cdot q_6 \cdot q_7$	$2^{61.68}$
7	2^7	2^6	$q_1 \cdot q_2$	$2^{5.52\dagger}$
23	2^{23}	2^{22}	$q_1 \cdot q_2 \cdot q_6$	$2^{21.49\dagger}$
38	2^{38}	2^{37}	$q_1 \cdot q_4 \cdot q_5 \cdot q_6$	$2^{36.5\dagger}$
55	2^{55}	2^{54}	$q_1 \cdot q_2 \cdot q_5 \cdot q_6 \cdot q_7$	$2^{53.49\dagger}$
63	2^{63}	2^{62}	$q_1 \cdot q_2 \cdot q_4 \cdot q_5 \cdot q_6 \cdot q_7$	$2^{61.5\dagger}$

* This complexity is obtained by GMP and higher-order differential attack.

* $(q_1, q_2, q_3, q_4, q_5, q_6, q_7) = (3, 5, 17, 257, 641, 65537, 6700417)$ are factors of $2^{64} - 1$.

† These distinguishers are constructed by merging sets. The set of monomials with non-zero exponents that are divisible by $|G|$ consists of $x^{|G|}$ and $x^{2|G|}$.

Table 8: Effective integral distinguishers found on Chaghri

Step	δ_r	Comp. of HD-ID	$ G $ *	Comp. of IPM-ID
8	17	2^{18}	$r_1 \cdot r_2 \cdot r_4$	$2^{17.39}$
15	35	2^{36}	$r_5 \cdot r_6$	$2^{35.81}$
17	40	2^{41}	$r_2 \cdot r_3 \cdot r_4 \cdot r_6$	$2^{40.88}$
17	40	2^{41}	$r_1 \cdot r_2 \cdot r_3 \cdot r_4 \cdot r_5$	$2^{40.88}$
24	58	2^{59}	$r_2 \cdot r_3 \cdot r_4 \cdot r_5 \cdot r_6$	$2^{58.97*}$

* $(r_1, r_2, r_3, r_4, r_5, r_6) = (7, 73, 127, 337, 92737, 649657)$ are factors of $2^{63} - 1$.

* This distinguisher is constructed by merging sets. The monomial with non-zero exponent that is divisible by $|G|$ found in 24-th step is $x^{2|G|}$.

 Table 9: Summation table of $\mathbb{X} := \{0, \alpha, \alpha^6, \alpha^{11}\}$.

$x \backslash j$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
α	1	α^1	α^2	α^3	α^4	α^5	α^6	α^7	α^8	α^9	α^{10}	α^{11}	α^{12}	α^{13}	α^{14}
α^6	1	α^6	α^{12}	α^3	α^9	1	α^6	α^{12}	α^3	α^9	1	α^6	α^{12}	α^3	α^9
α^{11}	1	α^{11}	α^7	α^3	α^{14}	α^{10}	α^6	α^2	α^{13}	α^9	α^5	α	α^{12}	α^8	α^4
$\bigoplus x^j$	0	0	0	α^3	0	0	α^6	0	0	α^9	0	0	α^{12}	0	0